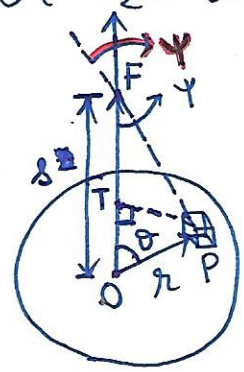
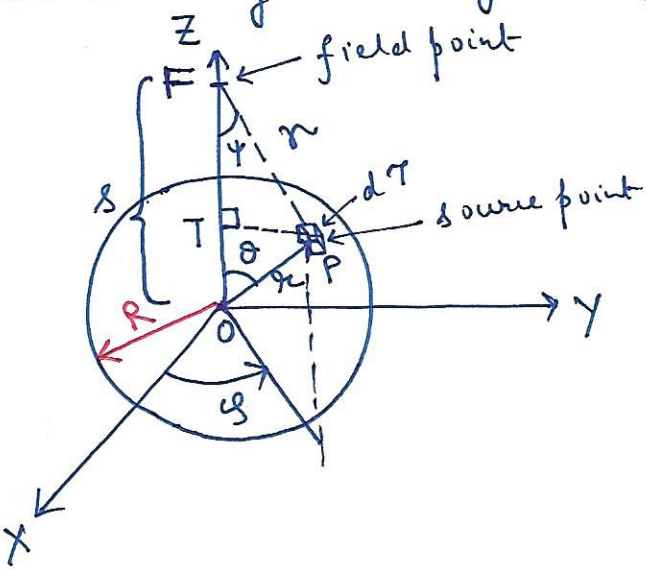
1. The charge density $\rho = \frac{\rho_0 r}{R}$ has a 'r' (radius) dependence only and thus the problem has a spherical symmetry. Moreover the electric field will be identical regardless of the direction of 's' you choose – hence we choose 's' along the z-direction (without any loss in generality).





Consider a source of elementary volume $d\tau$ located at r (w.r.t. O). Electric field due to this volume element of charge at F at a distance 's' from O is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{\rho\, d\tau}{r^2} \quad [r = \text{separation vector}]$$

In spherical co-ordinates, $d\tau = r^2 \sin\theta\, dr\, d\theta\, d\varphi$.

Using law of cosines, $r^2 = PF^2 = s^2 + r^2 - 2rs\cos\theta$. It is clear (from left fig) that due to the entire sphere, the effective field along z-axis would be $dE_{eff} = dE\cos\psi$

$$dE_{eff} = dE\,\frac{FT}{FP} = dE\left(\frac{OF - OT}{FP}\right) = dE\,\frac{s - r\cos\theta}{(s^2 + r^2 - 2rs\cos\theta)^{1/2}}$$

$$dE_{eff} = \frac{\rho_0 r}{4\pi\epsilon_0 R} \iiint \frac{r^2 \sin\theta\, dr\, d\theta\, d\varphi\,(s - r\cos\theta)}{(s^2 + r^2 - 2rs\cos\theta)^{3/2}}$$

$$E_{eff} = \int dE_{eff} = \frac{\rho_0}{4\pi\epsilon_0 R}\int_{r=0}^{R} dr \int_{\theta=0}^{\pi} d\theta \underbrace{\int_{\varphi=0}^{2\pi} d\varphi}_{(2\pi)} \frac{r^3 (s - r\cos\theta)\sin\theta}{(s^2+r^2-2rs\cos\theta)^{3/2}} \quad \text{—— (1)}$$

The φ- integral yields 2π (const). The other two integrals r and θ need to be calculated. We would calculate the θ- integral first (where r is taken to be constant).

$$E_{eff} = \frac{2\pi\rho_0}{4\pi\epsilon_0 R}\int_0^R r^3 dr \int_{\theta=0}^{\pi} \frac{(s - r\cos\theta)\sin\theta\, d\theta}{(s^2+r^2-2rs\cos\theta)^{3/2}}$$

. To solve θ-part,

let $u = \cos\theta$, $du = -\sin\theta\, d\theta$, $\theta = 0 \Rightarrow u = +1$, $\theta = \pi \Rightarrow u = -1$

$$I_\theta = \int_{u=-1}^{+1} \frac{(s - ru)\, du}{(s^2+r^2-2rsu)^{3/2}}$$

$$\boxed{E_{eff} = \frac{\rho_0}{2\epsilon_0 R}\int_0^R r^3 dr \int_{-1}^{+1} \frac{(s-ru)\, du}{(s^2+r^2-2sru)^{3/2}}} \quad \text{—— (2)}$$

Due to this special property of u- integral both cases of $s < R$ and $s > R$ need to be considered

$$\boxed{\begin{array}{ll} \frac{2}{s^2} & \text{for } s > r \\ 0 & \text{for } s < r \end{array}} \quad \text{—— (3)}$$

s: field point
r: source location

<u>Field inside sphere</u> <u>$(s < R)$</u>:

Note that r variable corresponds to position of source charge. Break the r integral into: i) $r: 0 \to s$, corresponding to the case $s > r$ in the u-integral (eqn 3), and ii) $r: s \to R$, corresponding to the case $s < r$ in the u-integral (eqn 3)

$$E_{eff} = \frac{\rho_0}{2\epsilon_0 R}\left[\int_{r=0}^{s} r^3 dr \times \left(\frac{2}{s^2}\right) + \int_{r=s}^{R} r^3 dr \times (0)\right] = E_{eff,in}$$

$$E_{eff,\,in} = \frac{\rho}{2\epsilon_0 R}\int_{r=0}^{s} r^3\,dr \times \frac{2}{s^2} = \frac{2\rho}{2\epsilon_0 R s^2}\,\frac{s^4}{4} = \frac{\rho}{4\epsilon_0 R}\,s^2$$

$$\boxed{\vec{E}_{eff,\,in} = \frac{\rho}{4\epsilon_0 R}\,s^2\,\hat{s}}$$

<u>Field outside sphere:</u> $(s > R)$

For points outside the sphere $(s>R)$, it is clear that $r < s$ always, so the u-integral is always $\frac{2}{s^2}$.

(2) reduces to

$$E_{eff,\,out} = \frac{\rho_0}{2\epsilon_0 R}\int_{r=0}^{R} r^3\,dr \cdot \frac{2}{s^2} \quad [r = \text{location of source}]$$

$$= \frac{\rho_0}{2\epsilon_0 R}\,\frac{2}{s^2}\,\frac{R^4}{4} = \frac{\rho_0 R^3}{4\epsilon_0}\cdot\frac{1}{s^2}$$

$$\boxed{\vec{E}_{eff,\,out} = \frac{\rho_0 R^3}{4\epsilon_0}\,\frac{1}{s^2}\,\hat{s}}$$

Optional (continuation of Q1)

$$I = \int_{u=-1}^{+1} \frac{(s - ru)\,du}{(s^2 + r^2 - 2rsu)^{3/2}}$$ can be solved by method of partial fractions.

$$I = \frac{1}{2s} \int_{-1}^{+1} \frac{(2s^2 - 2urs)\,du}{(s^2 + r^2 - 2rsu)^{3/2}}$$

$$= \frac{1}{2s} \int \left[\frac{(s^2 + r^2 - 2rsu) + (s^2 - r^2)}{(s^2 + r^2 - 2rsu)^{3/2}} \right] du$$

$$= \frac{1}{2s} \int_{-1}^{+1} \frac{du}{(s^2 + r^2 - 2rsu)^{1/2}} + \frac{1}{2s} (s^2 - r^2) \int_{-1}^{+1} \frac{du}{(s^2 + r^2 - 2rsu)^{3/2}}$$

$$\cancel{= \frac{1}{2s} \quad \frac{}{(s^2 + r^2 - 2rsu)}}$$

$$= \frac{1}{2s} \left. \frac{(s^2 + r^2 - 2rsu)^{1/2}}{\left(+\frac{1}{2} \times -2rs\right)} \right|_{-1}^{+1} + \frac{1}{2s} \frac{(s^2 - r^2)}{\left(-\frac{1}{2} \times -2rs\right)} \left. \frac{1}{(s^2 + r^2 - 2rsu)^{1/2}} \right|_{-1}^{+1}$$

$$= -\frac{1}{2rs^2} \left[\sqrt{(r-s)^2} - \sqrt{(r+s)^2} \right] + \frac{1 \times (s^2 - r^2)}{2rs^2} \left[\frac{1}{\sqrt{(r-s)^2}} - \frac{1}{\sqrt{(r+s)^2}} \right]$$

$$= -\frac{1}{2rs^2} \left[|r-s| - |r+s| \right] + \left(\frac{s^2 - r^2}{2rs^2} \right) \left[\frac{1}{|r-s|} - \frac{1}{|r+s|} \right]$$

For $s > r$, $|r-s| = s - r$, $|r+s| = r+s$

$$\frac{1}{|r-s|} = \frac{1}{s-r}, \quad \frac{1}{|r+s|} = \frac{1}{r+s}$$

$$I = -\frac{1}{2rs^2}\left[\cancel{s}-r-r-\cancel{s}\right] + \frac{s^2-r^2}{2rs^2}\left[\frac{1}{s-r} - \frac{1}{s+r}\right]$$

$$= \frac{1}{s^2} + \frac{s^2-r^2}{2rs^2}\left(\frac{\cancel{s}+r-\cancel{s}+r}{(s^2-r^2)}\right) = \frac{1}{s^2} + \frac{1}{s^2} = \frac{2}{s^2}$$

For $s < r$, $|r-s| = r-s$, $\frac{1}{|r-s|} = \frac{1}{r-s}$

$$I = -\frac{1}{2s^2r}\left[\cancel{r}-s-\cancel{r}-s\right] + \frac{s^2-r^2}{2rs^2}\left(\frac{1}{r-s} - \frac{1}{r+s}\right)$$

$$= \frac{1}{rs} + \frac{s^2-r^2}{2rs^2}\,\frac{\cancel{r}+s-\cancel{r}+s}{(r^2-s^2)} = \frac{1}{rs} - \frac{1}{rs} = 0$$

<u>Proved</u>

2 (a) $+q$ $\qquad\qquad$ $-2q$

$(x=2,\ y=-1,\ z=4)$ $\qquad$ $(x=-3,\ y=0,\ z=1)$

In this case the charge density is given by

$$\rho(\vec{r}) = \rho(x,y,z) = q\,\delta(x-2)\,\delta(y+1)\,\delta(z+4) - 2q\,\delta(x+3)\,\delta(y)\,\delta(z-1)$$

Check: take the volume integral on both sides of the entire space. This gives

$$\int_{\substack{\text{entire}\\\text{space}}} \rho(\vec{r})\, d^3r = q\int_{-\infty}^{\infty}\delta(x-2)\,dx \int_{-\infty}^{\infty}\delta(y+1)\,dy \int_{-\infty}^{\infty}\delta(z+4)\,dz - 2q\int_{-\infty}^{+\infty}\delta(x+3)\,dx \int_{-\infty}^{+\infty}\delta(y)\,dy \int_{-\infty}^{+\infty}\delta(z-1)\,dz$$

$$= q\,(1)(1)(1) - 2q\,(1)(1)(1)$$

$$= -q \quad \text{(net charge)}$$

(b) Suppose $\rho(\vec{r}) = \rho(r,\theta,\varphi) = Q\,a\,\delta(r-R)$. The factor '$a$' appears because we are now considering non-cartesian co-ordinates. $\delta(r-R)$ ensures that Q is distributed over the surface of spherical shell $r=R$.

Take volume integral on both sides over entire space.

$$\int_{(\infty)} \rho(\vec{r})\, d\tau = Q\,a\int_{(\infty)} \delta(r-R)\, d\tau \Rightarrow Q = Q\,a\int_{(\infty)} \delta(r-R)\, d\tau$$

Taking $d\tau = r^2 \sin\theta\, dr\, d\theta\, d\varphi$ in spherical co-ordinates

$$Q = Q\,a \int_{r=0}^{\infty} r^2\,\delta(r-R)\,dr \underbrace{\int_{\theta=0}^{\pi} \sin\theta\,d\theta \int_{\varphi=0}^{2\pi} d\varphi}_{2 \times 2\pi}$$

$$= 4\pi Q\,a \int_{0}^{\infty} r^2\,\delta(r-R)\,dr = 4\pi\,Q\,a\,R^2$$

$$\left[\because \int_{-\infty}^{+\infty} f(x)\,\delta(x-a) = f(a)\right]$$

$$\therefore\ Q = 4\pi Q\,a\,R^2$$

$$\Rightarrow \boxed{a = \frac{1}{4\pi R^2}}$$

This gives $\rho(\vec{r}) = \rho(r, \theta, \varphi)$

$$\boxed{\rho(\vec{r}) = \frac{Q}{4\pi R^2}\,\delta(r-R)}$$

[Note that units of delta function are always one over dimension of the argument of delta function]